

Scratch — Variables Intro

Let's learn how to use variables!

Grade 2 Coding



What is Coding?

Coding means giving instructions to a computer.

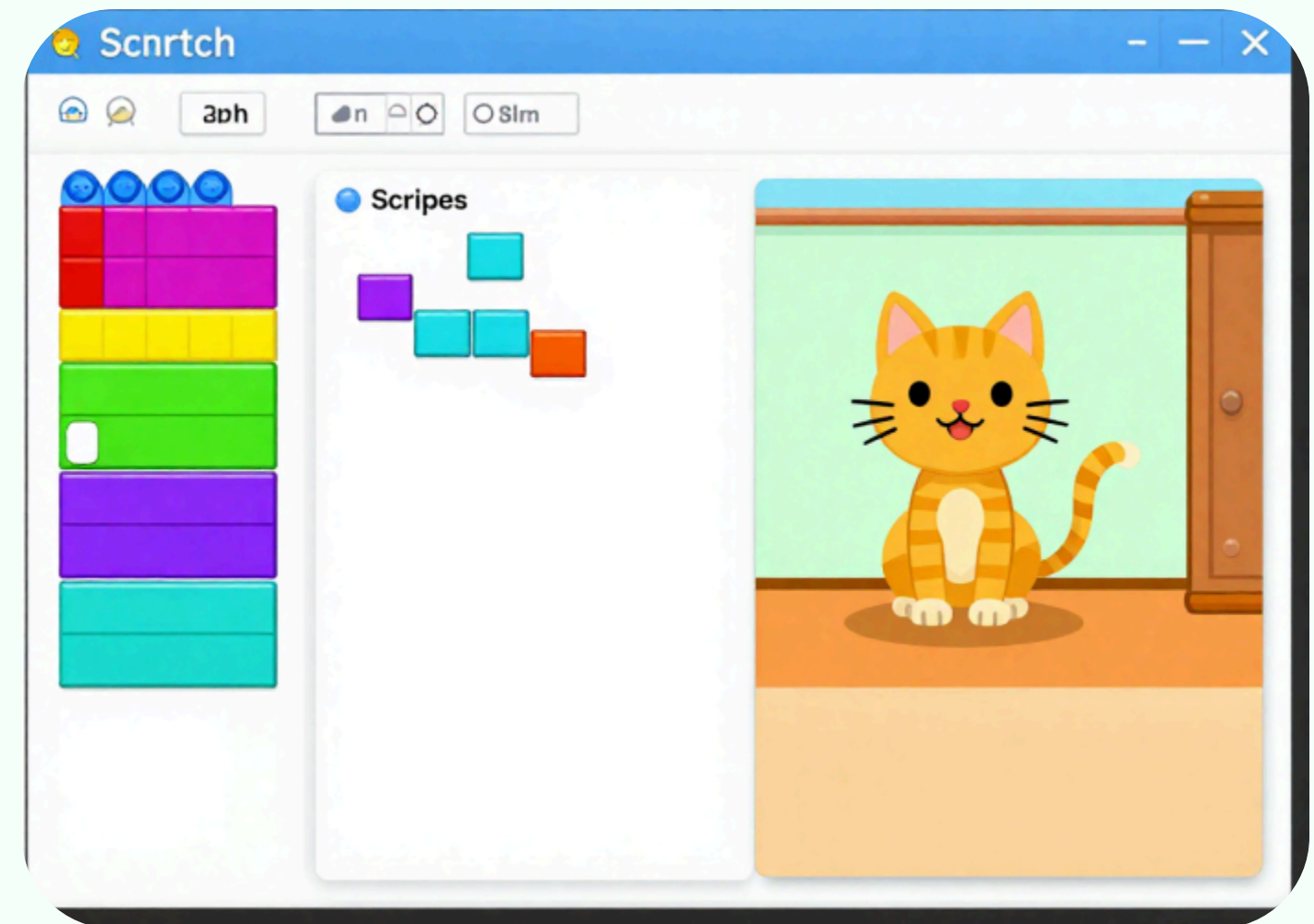
- Coding tells a computer what to do
- It follows your steps one by one
- Scratch is a fun way to learn coding!



Meet Scratch

Scratch is a colorful coding tool made just for kids!

- Drag and snap blocks together
- Make games, stories, and animations
- No typing needed — just click & drag!



What is a Variable?

A variable is like a little box where we store information!

Examples of variables:

- Score in a game
- Number of lives
- Player name

The box has a label (name) and something inside (value).



Why Do We Use Variables?

Variables help our program remember!

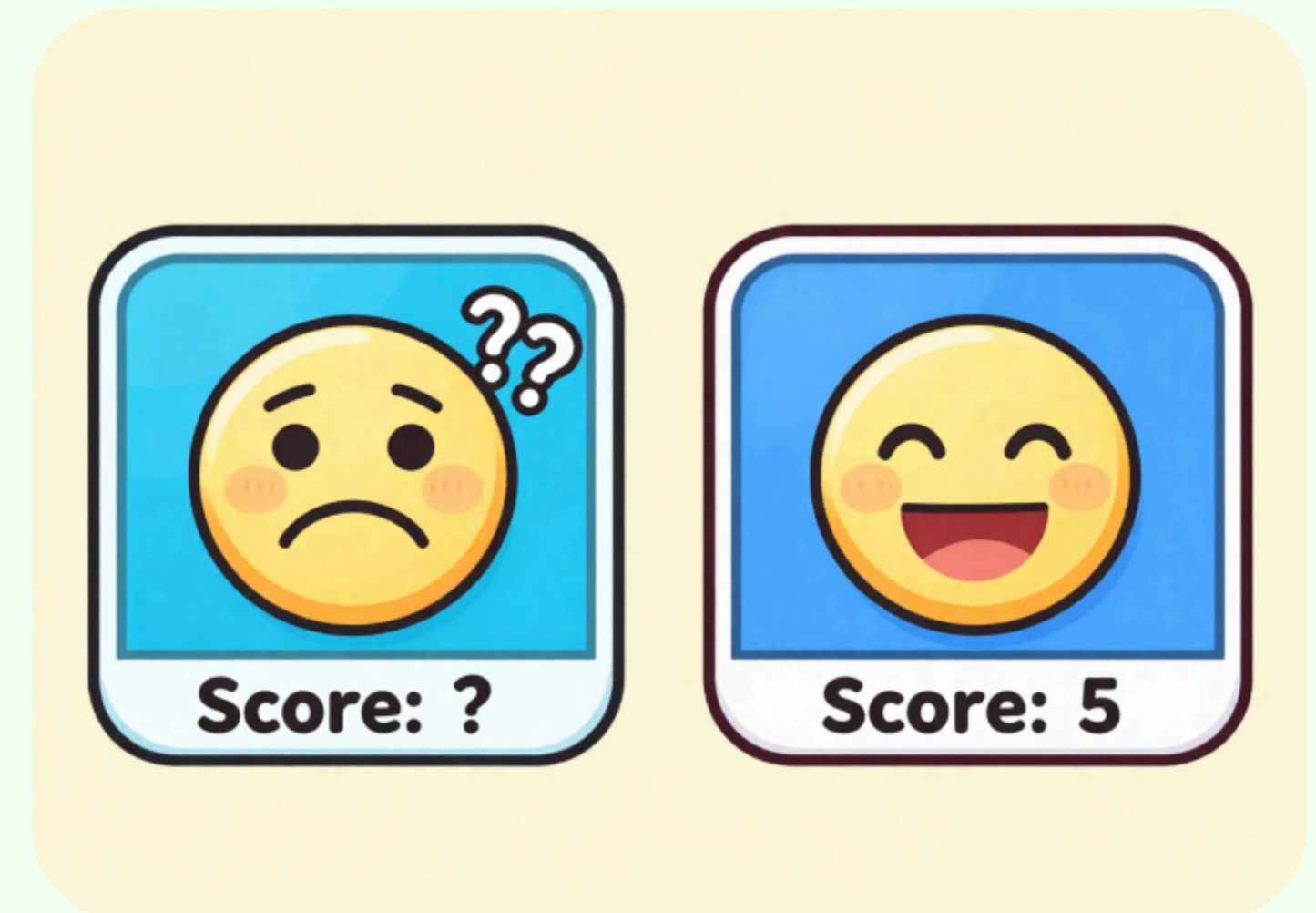
- To keep track of scores or counts
- To remember things while playing
- To change things as the game goes on

Without variables:

The computer forgets everything!

With variables:

The score goes up when you win!



How to Create a Variable in Scratch

Follow these steps:

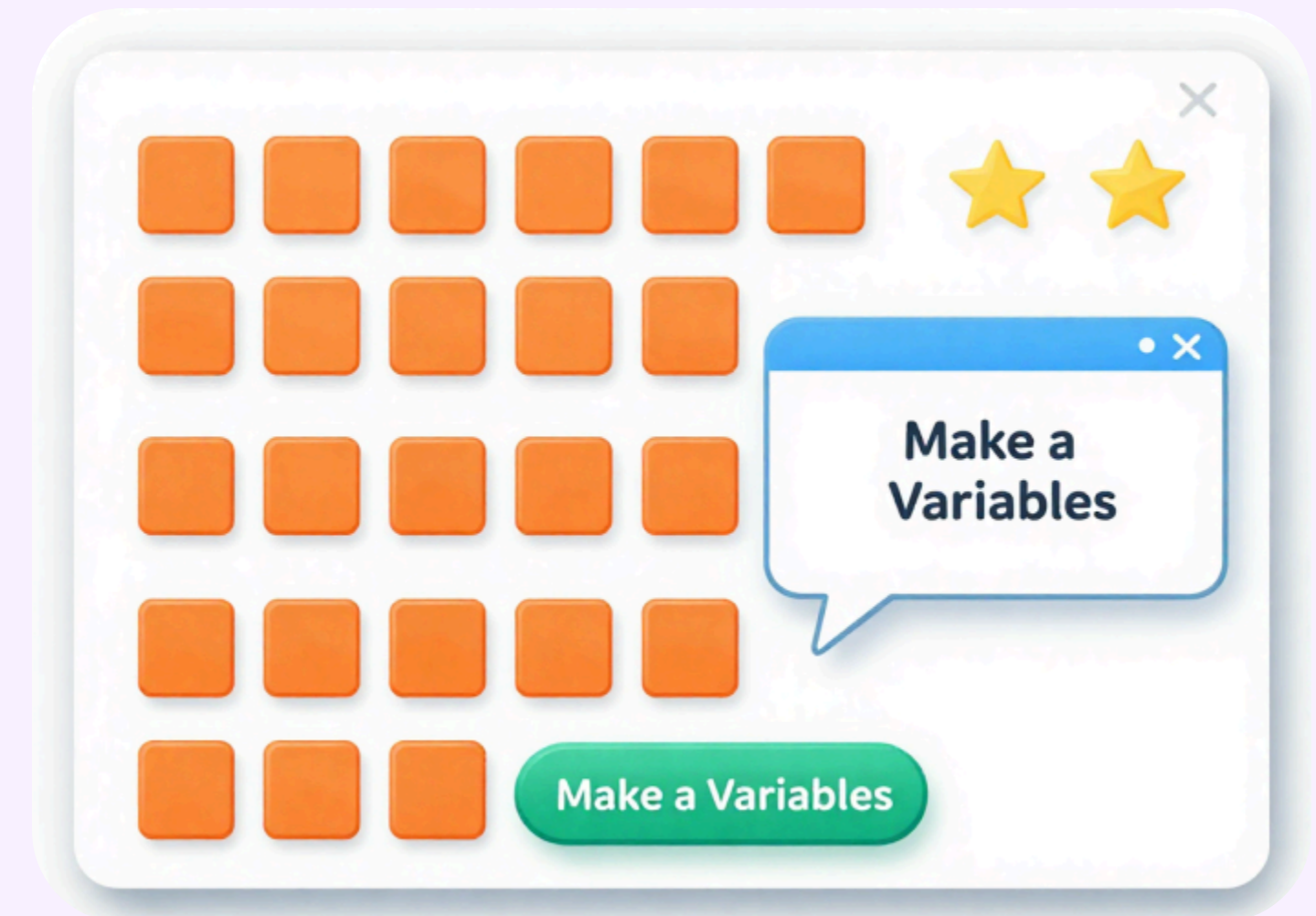
Step 1: Click the 'Variables' category
(orange blocks)

Step 2: Click 'Make a Variable'

Step 3: Type a name for your variable
(like 'score')

Step 4: Click OK

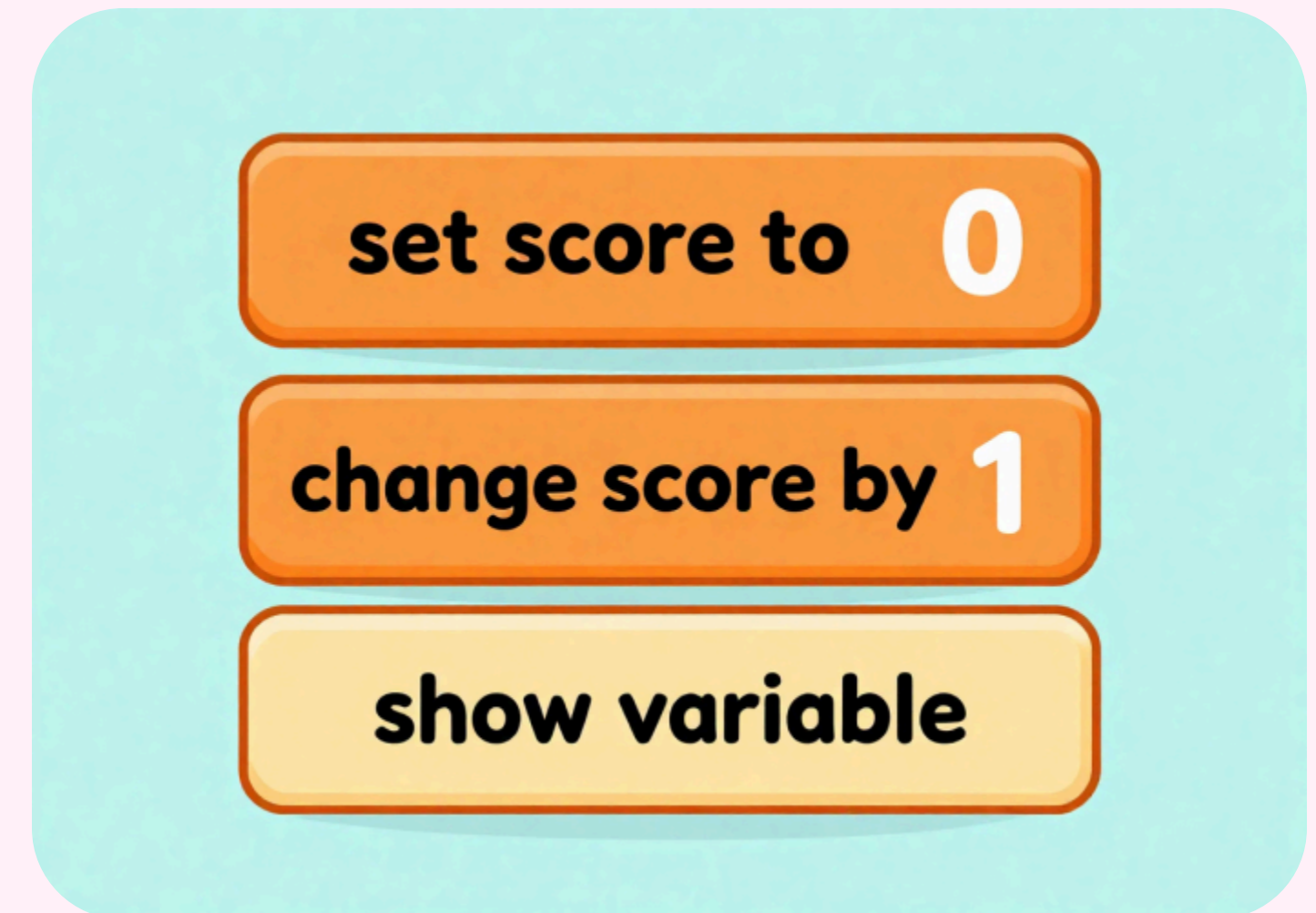
Your new variable appears on the stage!



Using Variables in Scratch

Here are the main variable blocks:

- Set score to 0
(gives the variable a starting value)
- Change score by 1
(adds 1 to the score)
- Show variable
(displays it on the stage)
- Hide variable
(hides it from the stage)



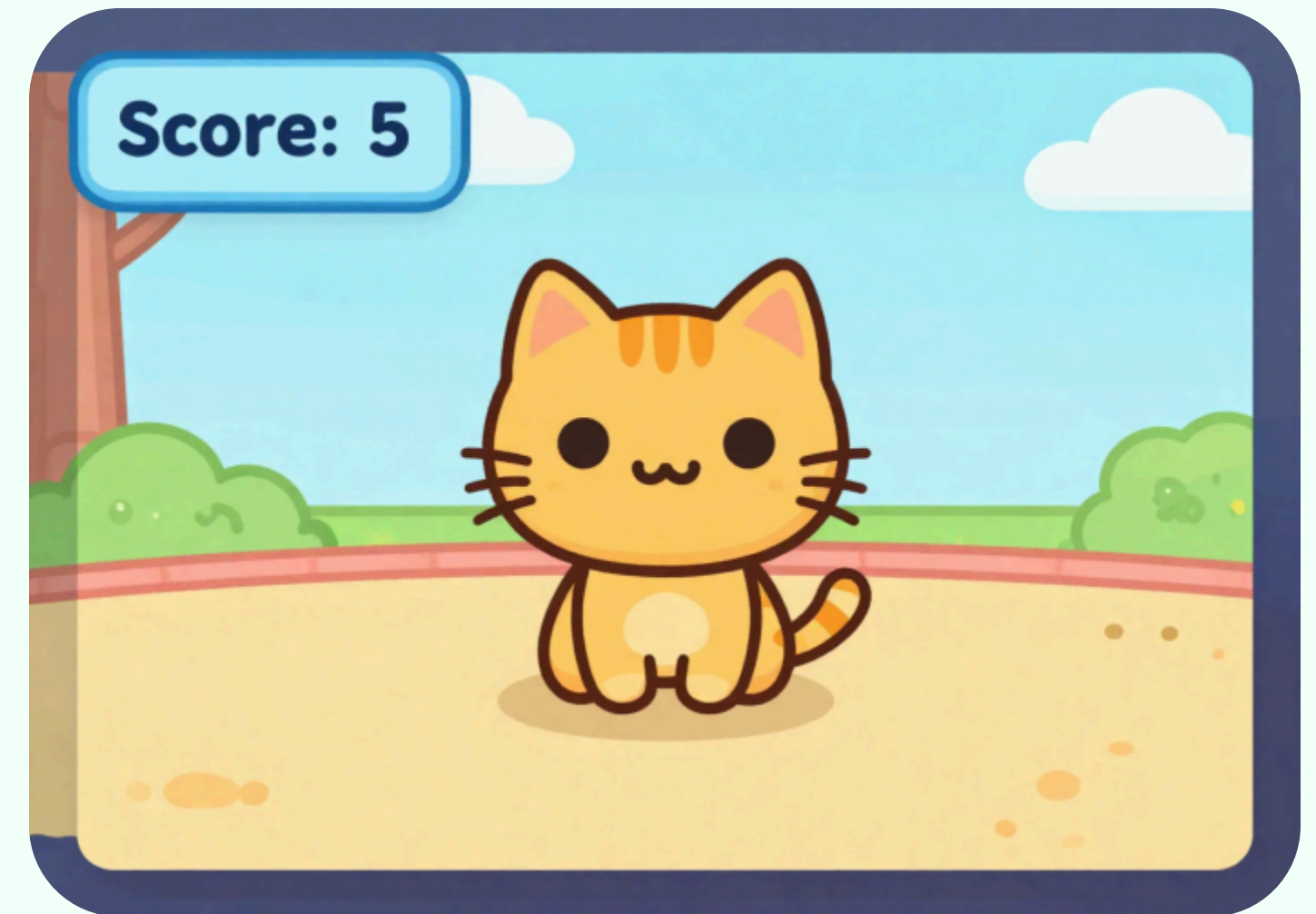
Example: Score Counter

Let's make a score counter game!

How it works:

- Set score to 0 at the start
- When the sprite is clicked, change score by 1
- The score shows on the stage!

Try clicking the sprite many times and watch the score go up!



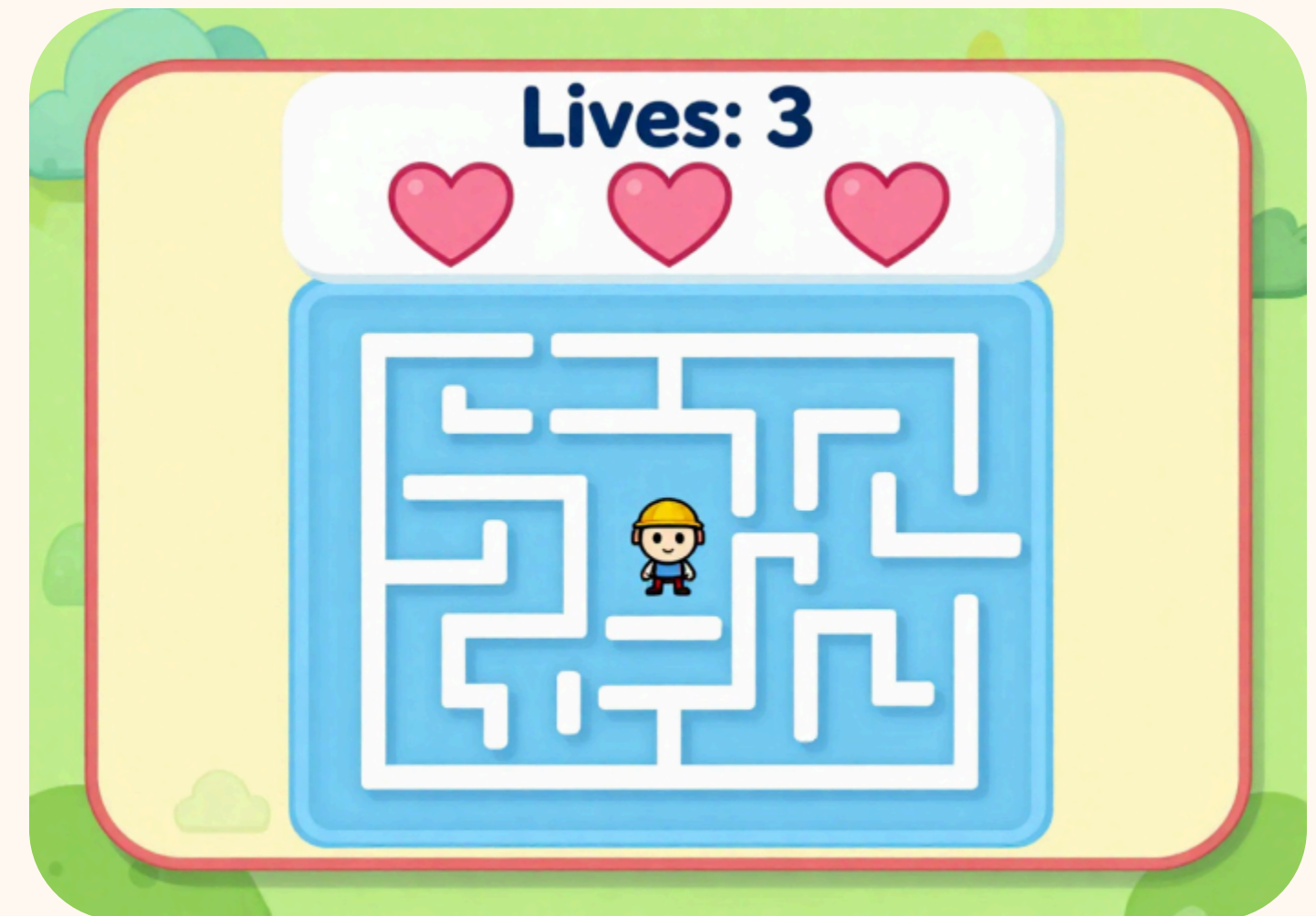
Example: Lives Tracker

Now let's track lives in a game!

How it works:

- Set lives to 3 at the start
- When something bad happens, change lives by -1
- If lives = 0, game over!

This way the computer remembers how many chances you have left.



Let's Try It!

Your turn! Make your own variable!

Think about:

- What game will you make?
- What will your variable track?
(score, lives, time, coins?)
- What will you name it?

Open Scratch and try creating a variable right now!



Recap: What Did We Learn?

Great job! Here's what we learned:

- ✓ A variable is a box that stores information
- ✓ Variables help us remember and change things in our programs
- ✓ We can create variables in Scratch using the orange blocks
- ✓ We can set, change, show, and hide variables!



Quiz Time! Question 1

What is a variable?

- A) A box to store information
- B) A type of game
- C) A color in Scratch



Quiz Time! Question 2

Which of these can be a variable?

A) Score

B) Player name

C) Both A and B



Answers

Question 1: What is a variable?

- ✓ Answer: A) A box to store information
A variable holds data like scores, names, or lives.

Question 2: Which can be a variable?

- ✓ Answer: C) Both A and B
Score AND player name can both be stored in variables!



Next Steps

Keep exploring and practising!

- Try adding variables in your own Scratch projects
- Make a game with a score counter
- Ask a teacher or parent for help
- Practice makes you a better coder!

You're doing amazing — keep going!



Thank You!

Thank you for learning about
variables with Scratch!

Now you know how to store and
change information in your programs.

Keep coding and have fun!

